

Overview of Regularization machine learning

Subject: Regularization machine learning

1.

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

2.

$R(w) = \frac{1}{2} \sum \sigma(W)^2$ where $\sigma(W)$ is the eigenvalues in the singular value decomposition of W .

3.

Group sparsity without overlaps Groups of features can be regularized by a sparsity constraint, which can be useful for expressing certain prior knowledge into an optimization problem. In the case of a linear model with non-overlapping known groups, a regularizer can be defined:

4.

$w_T = \frac{\gamma}{\sum_{i=0}^{T-1} (I - \gamma X^T X)^i} X^T Y$

5.

$\text{prox}_{\lambda, R, g}(w_g) = \begin{cases} (1 - \lambda \|w_g\|_2) w_g, & \text{if } \|w_g\|_2 > \lambda \\ 0, & \text{if } \|w_g\|_2 \leq \lambda \end{cases}$

6.

$\min_{w \in \mathbb{R}^{p \times 1}} \|X^T w - Y\|^2 + \lambda (\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2), \alpha \in [0, 1]$

7.

Without bounds on the complexity of the function space (formally, the reproducing kernel Hilbert space) available, a model will be learned that incurs zero loss on the surrogate empirical error. If

measurements (e.g. of x_i) were made with noise, this model may suffer from overfitting and display poor expected error. Regularization introduces a penalty for exploring certain regions of the function space used to build the model, which can improve generalization.

8.

Regularization can be motivated as a technique to improve the generalizability of a learned model. The goal of this learning problem is to find a function that fits or predicts the outcome (label) that minimizes the expected error over all possible inputs and labels. The expected error of a function f_n is:

9.

When labels are more expensive to gather than input examples, semi-supervised learning can be useful. Regularizers have been designed to guide learning algorithms to learn models that respect the structure of unsupervised training samples. If a symmetric weight matrix W is given, a regularizer can be defined:

10.

$$w_0 = 0 \quad w_{t+1} = (I - \gamma_n X^T X) w_t + \gamma_n X^T Y$$

11.

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

12.

$$R(f_1 \dots f_T) = \sum_{t,s=1, t \neq s}^T \|f_t - f_s\|_{M_{ts}}^2$$

for a given symmetric similarity matrix M .

13.

Explicit regularization is regularization whenever one explicitly adds a term to the optimization problem. These terms could be priors, penalties, or constraints. Explicit regularization is commonly employed with ill-posed optimization problems. The regularization term, or penalty, imposes a cost on the optimization function to make the optimal solution unique. Implicit regularization is all other forms of regularization. This includes, for example, early stopping, using a robust loss function, and discarding outliers. Implicit regularization is essentially ubiquitous in modern machine learning approaches, including stochastic gradient descent for training deep neural networks, and ensemble methods (such as random forests and gradient boosted trees). In explicit regularization, independent of

the problem or model, there is always a data term, that corresponds to a likelihood of the measurement, and a regularization term that corresponds to a prior. By combining both using Bayesian statistics, one can compute a posterior, that includes both information sources and therefore stabilizes the estimation process. By trading off both objectives, one chooses to be more aligned to the data or to enforce regularization (to prevent overfitting). There is a whole research branch dealing with all possible regularizations. In practice, one usually tries a specific regularization and then figures out the probability density that corresponds to that regularization to justify the choice. It can also be physically motivated by common sense or intuition. In machine learning, the data term corresponds to the training data and the regularization is either the choice of the model or modifications to the algorithm. It is always intended to reduce the generalization error, i.e. the error score with the trained model on the evaluation set (testing data) and not the training data. One of the earliest uses of regularization is Tikhonov regularization (ridge regression), related to the method of least squares.

14.

For each w_g , $\bar{w}_g$ is defined as the vector such that the restriction of $\bar{w}_g$ to the group g equals w_g and all other entries of $\bar{w}_g$ are zero. The regularizer finds the optimal disintegration of w into parts. It can be viewed as duplicating all elements that exist in multiple groups. Learning problems with this regularizer can also be solved with the proximal method with a complication. The proximal operator cannot be computed in closed form, but can be effectively solved iteratively, inducing an inner iteration within the proximal method iteration.